Kyle D. Singer

Education

Ph.D. in Computer Science, August 2017 - May 2023
Washington University in St. Louis
PhD Advisor: I-Ting Angelina Lee

- *Dissertation:* An Efficient Task-Parallel Platform for Interactive Applications

M.S. in Computer Science August 2017 - December 2022
Washington University in St. Louis GPA: 4.0/4.0

B.S. in Computer Science August 2010 - May 2013
University of Evansville GPA: 3.95/4.0
Minor: Mathematics
Summa Cum Laude

Publications

Kyle Singer, Tao B. Schardl, Kunal Agrawal. Waste-Efficient Work Stealing. In Proceedings of the 31st ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (PPoPP'26). URL: https://doi.org/10.1145/3774934.3786452.

Aaron Handleman, **Kyle Singer**, Tao B. Schardl, I-Ting Angelina Lee. Towards Zero Spawn Overhead: Work Stealing Without Deques. In Proceedings of the 37th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA'25). URL: https://doi.org/10.1145/3694906.3743349.

Kyle Singer, Kunal Agrawal, I-Ting Angelina Lee. An Efficient Scheduler for Task-Parallel Interactive Applications. In Proceedings of the 35th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA'23). URL: https://doi.org/10.1145/3558481.3591092.

Stefan K Muller, **Kyle Singer**, Devyn Terra Keeney, Andrew Neth, Kunal Agrawal, I-Ting Angelina Lee, Umut A Acar. Responsive Parallelism with Synchronization. In Proceedings of the ACM on Programming Languages (PACMPL), Volume 7, Issue PLDI (PLDI'23). URL: https://doi.org/10.1145/3591249.

Kyle Singer. An Efficient Task-Parallel Platform for Interactive Applications. PhD Dissertation. Department of Computer Science and Engineering, Washington University in St. Louis, 2023. URL: https://www.proquest.com/dissertations-theses/efficient-task-parallel-platform-interactive/docview/2806338708/se-2.

Yifan Xu, **Kyle Singer**, I-Ting Angelina Lee. Parallel Race Detection with Futures. In Proceedings of the 25th Symposium on Principles and Practice of Parallel Programming (PPoPP'20). URL: https://doi.org/10.1145/3332466.3374536**.**

Kyle Singer, Noah Goldstein, Stefan Muller, Kunal Agrawal, I-Ting Angelina Lee, Umut A. Acar. Priority Scheduling for Interactive Applications. In Proceedings of the 32nd Symposium on Parallelism in Algorithms and Architectures (SPAA'20). URL: https://doi.org/10.1145/3350755.3400236.

Stefan K. Muller,* **Kyle Singer**,* Noah Goldstein, Umut A. Acar, Kunal Agrawal, and I-Ting Angelina Lee. Responsive Parallelism with Futures and State. In Proceedings of the 41st Conference on Programming Language Design and Implementation (PLDI'20). URL: https://doi.org/10.1145/3385412.3386013.
***The first two authors contributed equally to this work.**

Kyle Singer, Kunal Agrawal, I-Ting Angelina Lee. Scheduling I/O Latency-Hiding Futures in Task-Parallel Platforms. In Proceedings of the 1st Symposium on Algorithmic Principles of Computer Systems (APoCS'20). URL: https://doi.org/10.1137/1.9781611976021.11.

Kyle Singer, Yifan Xu, and I-Ting Angelina Lee. 2019. Proactive work stealing for futures. In Proceedings of the 24th Symposium on Principles and Practice of Parallel Programming (PPoPP '19). URL: https://doi.org/10.1145/3293883.3295735.

Patents

Bruce Ianni, Davyeon Ross, Justin Bennett, Mike Ciholas, Herb Hollinger, **Kyle Singer**, and Dirk VanVorst. 2018. Transaction scheduling system for a wireless data communication network. US Patent Publication No. US9858451B2.

Research Experience

Postdoctoral Associate - SuperTech Research Group July 2023 - August 2026
Massachusetts Institute of Technology Cambridge, MA
Advisor: Charles E. Leiserson

Research Assistant - Parallel Computing Technology Group August 2017 - July 2023
Washington University in St. Louis St. Louis, MO
Advisor: I-Ting Angelina Lee

Teaching Experience

Computer Science Teaching Assistant January 2020 - May 2020
CSE539S: Concepts in Multicore Computing St. Louis, MO
Washington University in St. Louis
- Gave several lectures, assisted students in office hours, and graded assignments

English Teacher September 2013 - February 2014
Multiple Classes Chungju-si, Chungcheongbuk-do, South Korea
Top Academy
- Instructed elementary and middle school students in English reading, writing, speaking, and listening at a private after-school academy.

Computer Science Tutor March 2013 - May 2013
CS 215: The Fundamentals of Programming II Evansville, IN
University of Evansville
- Tutored undergraduate students in CS 215, the Fundamentals of Programming II.

Physics Teacher's Assistant August 2011 - May 2013
Physics 121 Lab, Physics 122 Lab Evansville, IN
University of Evansville

- Aided undergraduate students in comprehending and applying principles of Physics in a lab setting.
- Graded undergraduate physics lab papers submitted to the University of Evansville's Physics department.

Computer Science Grader January 2012 - May 2012
CS 210: The Fundamentals of Programming I Evansville, IN
University of Evansville

- Graded undergraduate programming projects submitted.

Other Work Experience

Software Engineer Intern (Part-time) August 2021 – January 2023
Neural Magic Boston, MA (Remote)

- Developed *offline* and *online* scheduling algorithms to improve performance of machine learning algorithms in a software system that provides GPU-class performance on CPU hardware.

Software Engineer Intern, Systems and Infrastructure (PhD) May 2021 – August 2021
Facebook Boston, MA (Remote)

- Worked to enhance security for services that utilize an internal service management platform.

Software Engineer Intern January 2021 – May 2021
Neural Magic Boston, MA (Remote)

- Developed *online* scheduling algorithms to improve performance of machine learning algorithms on multi-socket computers.

Software Engineer September 2014 – June 2017
Ciholas Inc. Newburgh, IN

- Worked with a team to design an ultra-wideband radio network that can track locations with high precision.
- The primary developer on the dwusb_gui and the subsequent cuwb_server desktop applications used both to perform location calculations as well as to control the behavior of devices on the radio network.
- Provided on-site customer support and setup of ultra-wideband networks at locations such as CES in Las Vegas and CES Asia in Shanghai, China.
- Wrote an Android library to allow clients to interface with a USB-connected embedded ultra-wideband device that generates location data.
- Wrote firmware for ultra-wideband devices for use with the dwusb_gui and cuwb_server applications.
- Designed and implemented a transaction scheduling scheme for collecting data to perform location calculations using the two-way ranging plus snoop algorithm.
- Wrote firmware to decode and play audio in the Speex format when triggered remotely via radio.

Intern Summer 2010-2011, Summer 2013
Magellan Health Services Maryland Heights, MO

- Wrote SQL for a system that automated moving data from files into databases.
- Eliminated unnecessary work by designing and implementing code to automate manual logging procedures.
- Consulted on task automation procedures.
- Tested software that interfaces with Microsoft SQL Server prior to deployment.

Talks

An Efficient Scheduler for Task-Parallel Interactive Applications

- Conference talk at SPAA'23 — June 2023

An Efficient Task-Parallel Platform for Interactive Applications

- Doctoral Dissertation Defense — April 2023

Priority Scheduling for Interactive Applications

- Conference talk at SPAA'20 — July 2020

Responsive Parallelism with Futures and State

- Conference talk at PLDI'20 — June 2020

Scheduling I/O Latency-Hiding Futures in Task-Parallel Platforms

- Conference talk at APoCS'20 — January 2020
- Seminar talk at Washington University in St. Louis — November 2019

Reduced I/O Latency with Futures

- Brief announcement at SPAA'19 — June 2019

Proactive Work Stealing for Futures

- Conference talk at PPoPP'19 — February 2019
- Seminar talk at Washington University in St. Louis — December 2018

Professional Services

Program Committee Member — 2026
SPAA'26 — Remote

- Reviewed paper submissions to the SPAA 2026 conference.

Artifact Evaluation Committee Member — 2025
ALENEX'26 — Remote

- Verified results reported in several papers submitted to the ALENEX 2026 conference.

Artifact Evaluation Committee Member — 2024
ALENEX'25 — Remote

- Verified results reported in several papers submitted to the ALENEX 2025 conference.

Paper Subreviewer — 2024
SPAA'24 — Remote

- Reviewed paper submission to the SPAA 2024 conference.

Paper Subreviewer — 2023
SPAA'23 — Remote

- Reviewed paper submission to the SPAA 2023 conference.

Artifact Evaluation - Reviewer — 2019

PPoPP’20 Remote

- Verified results reported in several papers submitted to the PPoPP 2020 conference.

High School Programming Contest Judge 2012
University of Evansville ACM Evansville, IN

- Verified contestants' submissions through the use of various test cases.

Volunteer Webelos Day Instructor 2011
University of Evansville ACM Evansville, IN

- Instructed elementary school students in the basics of computer science using MIT Scratch.

High School Programming Contest Problem Author 2011
University of Evansville ACM Evansville, IN

- Developed level appropriate problems for high school students to solve in a programming contest.

ACM Secretary September 2011 – May 2012
University of Evansville ACM Evansville, IN

- Assisted in the planning and organization of activities for the University of Evansville's ACM chapter.